UNIT II: Ignition & Fuel Supply Systems

Syllabus: Ignition system - coil & magneto, spark plug, distributor, electronic ignition; fuel system - carburettor, fuel pumps, fuel injection (mono-point, multi-point), unit injector, nozzle types, EFI, GDI, MPFI, DTS.

1. Ignition System - Introduction

- Ignition system supplies a high-voltage spark (10-30 kV) across the spark plug gap at the correct instant to ignite the compressed air-fuel mixture in SI (petrol) engines.

Functions

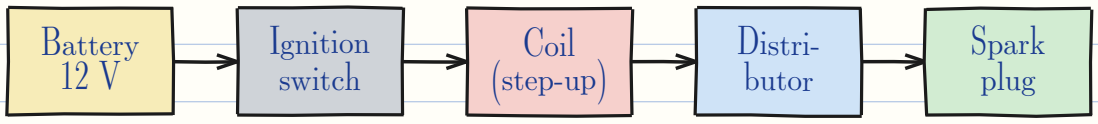
- Step up battery voltage (12 V) to a very high voltage.
- Distribute the spark to each cylinder in the firing order.
- Time the spark correctly (advance with speed & load).

Requirements of a good ignition system

- Strong spark at all speeds & loads (even at cranking).
- Accurate timing; reliable; long life; low maintenance; interference-free (RFI suppression).

Types of ignition systems

1. Battery (coil) ignition - conventional, with contact breaker & distributor.
2. Magneto ignition - self-generating, no battery; two-wheelers, racing, small engines.
3. Electronic ignition - transistorised (TCI), capacitive discharge (CDI), distributorless (DIS), coil-on-plug (COP).



Basic chain: 12 V -> 20,000 V -> spark in cylinder

Fig 2.1 Block diagram of ignition system

Spark energy

Minimum ignition energy ~0.2 mJ (stoichiometric); rich / cold start needs ~3 mJ. Spark lasts about 1-2 ms. Required voltage rises with gap, compression, rpm & lean mixture.

2. Battery (Coil) Ignition System

- Used in cars (older) with contact breaker + distributor. Two circuits: primary (low tension, LT) and secondary (high tension, HT).

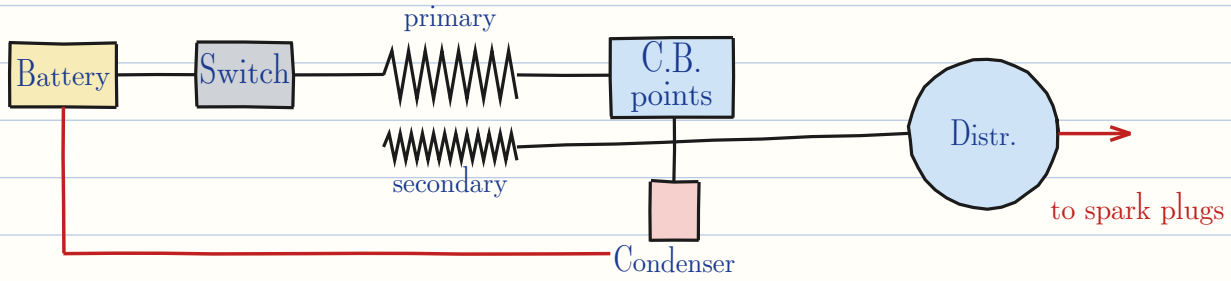


Fig 2.2 Battery coil ignition circuit

Components

- Battery (12 V), ignition switch, ballast resistor (reduces primary voltage at high speed; bypassed while cranking).
- Ignition coil (autotransformer), contact breaker (CB) points + cam, condenser (0.2 uF), distributor, spark plugs, HT leads.

Working

1. Switch ON, CB points closed: current ~3-4 A flows in the primary winding and builds a magnetic field around the iron core.
2. Cam opens the points: primary current is cut suddenly -> field collapses, cutting the many turns of the secondary winding.
3. EMF induced in secondary = 20-30 kV (turns ratio ~ 100:1) -> sent through the distributor rotor to the correct plug -> spark jumps the gap.
4. Condenser absorbs self-induced EMF (~250 V) in the primary, prevents arcing at points and makes the field collapse faster -> stronger spark.

Dwell angle

Angle through which the CB points remain closed (about 60 deg for 4-cyl; ~ 60% of cycle). Short dwell -> weak spark at high speed; long dwell -> points burn.

3. Ignition Coil & Contact Breaker

- The ignition coil is a pulse transformer: low turns on primary (200-300 turns of thick wire) + very many turns on secondary (20,000 - 25,000 turns of fine wire) wound on a laminated soft-iron core.

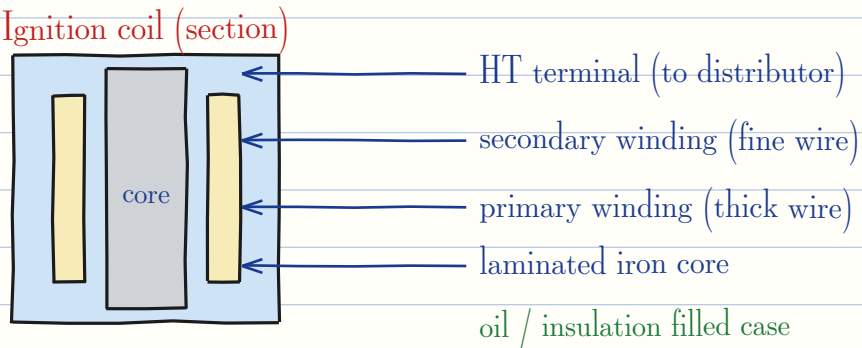


Fig 2.3 Ignition coil construction

Key points

- Turns ratio = N2 / N1 ~ 100:1 ; secondary voltage V2 = V1 x (N2/N1) [theory]; but real voltage depends on rate of flux collapse.
- Energy stored: E = 1/2 L I^2 (joules) - about 30-60 mJ.
- Coil is cased in oil / resin for insulation & cooling. Dry (epoxy) coils are used in modern cars.

Contact breaker (CB) unit

- Spring-loaded moving arm with tungsten contacts, fixed point, fibre rubbing block following the cam (lobes = number of cylinders).
- Point gap 0.35 - 0.45 mm; gap too small -> long dwell, weak spark at low speed; too large -> short dwell.
- Faults: burnt / pitted points, wrong gap, weak spring, faulty condenser (excess arcing), oil on points.

Fault	Effect
Wide point gap	Late timing / short dwell / misfire at high speed
Narrow gap	Early timing / long dwell / burnt points
Faulty condenser	Heavy arcing at points, weak or no spark

4. Magneto Ignition System

- A magneto generates its own current (no battery needed) by rotating magnets near coils - used on two-wheelers, scooters, lawn-mowers, aircraft & racing engines .
- Working principle: same as coil ignition, but the primary current is produced by electromagnetic induction from a rotating permanent magnet.

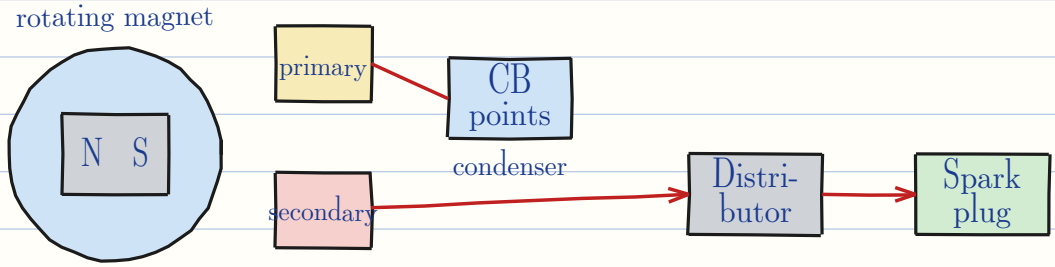


Fig 2.4 Magneto ignition (block view)

Types of magneto

- Armature-rotating type: coils rotate between fixed magnets (old). Magnet-rotating type: magnet rotates, coils fixed - common. Polar-inductor type: only iron cores rotate, magnet & coils fixed.
- Flywheel magneto - magnets in flywheel rim; coil outside; used in scooters, motorcycles (also with CDI).

Coil vs Magneto ignition

Point	Battery coil	Magneto
Battery	Required	Not required
Starting	Good (strong spark at low rpm)	Poor at cranking speed
Spark at high rpm	Weaker	Stronger
Weight / cost	Cheaper, easy repair	Costlier, compact
Use	Cars, buses, LCVs	Two-wheelers, racing, small engines

5. Spark Plug

- The spark plug provides the air gap where the high-voltage spark jumps and ignites the mixture. Screwed into the cylinder head.

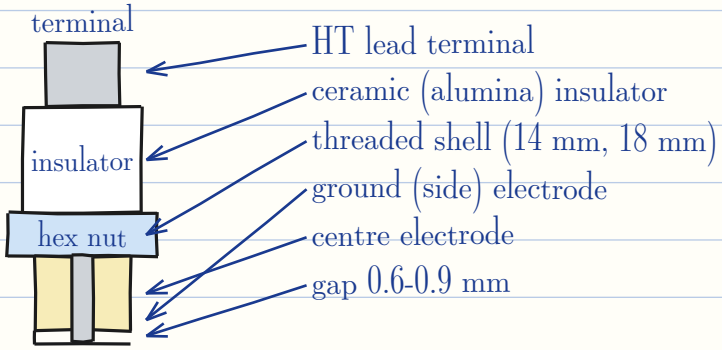


Fig 2.5 Spark plug construction

Parts & materials

- Centre electrode - nickel alloy / copper core / platinum / iridium (long life). Ground electrode welded to shell.
- Insulator - sintered alumina ceramic, withstands 30 kV and 1000 deg C; steel shell with thread; copper gasket seals.

Heat range (hot & cold plugs)

- Heat range = ability to transfer heat from firing tip to the head. Hot plug: long insulator nose, slow heat loss - for low-speed / light-duty engines. Cold plug: short nose, rapid heat loss - for high-speed / high-compression engines.
- Self-cleaning temperature 500 deg C (burns carbon); pre-ignition above 850 deg C.

Condition	Meaning of plug tip
Light brown / grey	Normal; correct heat range
Black, sooty	Rich mixture / plug too cold
White / blistered	Lean / overheating / plug too hot
Oily wet	Oil leaking past rings / valves

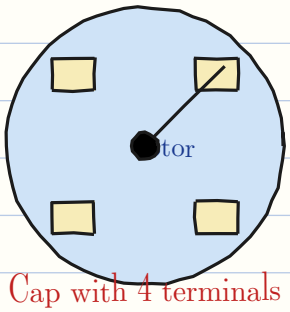
- Gap is set with feeler gauge (0.6 - 0.8 mm); replace every 20,000 - 40,000 km (platinum / iridium ~ 80,000 km).

6. Distributor & Ignition Timing

- The distributor (a) opens/closes primary circuit through CB points, (b) distributes HT to plugs in firing order, (c) advances timing automatically.

Parts

- Cap (bakelite, with central & outer terminals), rotor arm, CB points + cam, condenser, driven shaft (half crank speed), centrifugal advance weights, vacuum advance diaphragm, oil-felt for cam lubrication.



Cap with 4 terminals

Rotor sweeps past terminals in firing order 1-3-4-2

- small gap (0.3 mm) between rotor tip and terminal (spark jumps)

Centre terminal <- HT from coil

Fig 2.6 Distributor cap and rotor (top view)

Automatic spark advance

- Centrifugal advance: bob-weights fly out with speed, rotating cam forward against springs -> spark earlier at high rpm (flame needs more crank angle).
- Vacuum advance: diaphragm connected to inlet manifold vacuum (near throttle) moves CB plate -> spark advanced at part load (lean mixture burns slowly). Retards at full load.
- Octane selector - manual fine adjust of timing for fuel grade.

Ignition timing

Spark occurs 5-15 deg BTDC at idle, up to 30-40 deg BTDC at high speed. Too advanced -> knocking; too retarded -> loss of power, overheating. Checked with a stroboscopic timing light.

7. Electronic Ignition System

- Replaces mechanical CB points with a transistor / thyristor switch triggered by a sensor - no wear, no maintenance, stronger spark (up to 35-40 kV), better cold start & emission.

Types

- TCI (transistorised coil ignition) - pickup (inductive or Hall-effect sensor, or optical) sends pulse to an igniter module which switches the primary current.
- CDI (capacitor discharge ignition) - a capacitor is charged to 300-400 V by a DC-DC converter, then dumped through the coil primary by a thyristor (SCR) -> very fast rise time; used in motorcycles.
- DIS (distributorless / waste-spark) - one coil for two cylinders; both plugs fire together (one in compression, one in exhaust). No rotor or cap.
- COP (coil-on-plug) - individual coil mounted on each plug; no HT leads; ECU controls timing for each cylinder. Best energy; used in modern engines.

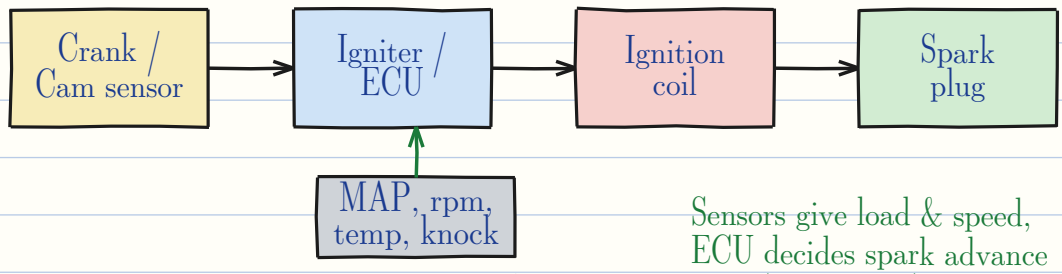


Fig 2.7 Electronic ignition (with ECU)

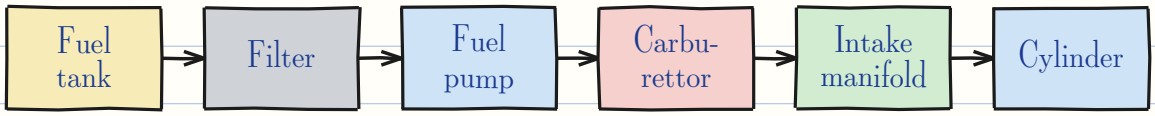
Advantages

- No contact wear / dwell setting; consistent high voltage at all speeds; programmed timing map (3-D map of load x rpm) instead of mechanical advance weights.
- Knock sensor retards timing automatically; allows higher compression ratio; lower emissions and better fuel economy.
- Hall-effect sensor: a rotating shutter (vane) cuts a magnetic field in a Hall IC producing a clean square-wave pulse.

8. Fuel System - Overview

- Purpose: store fuel, filter it, and deliver the correct quantity of fuel (in correct air-fuel ratio) to the engine at all speeds and loads.

Petrol engine - carburettor system



Air cleaner -> air also enters the carburettor
Return line brings excess fuel back to the tank (injection systems)

Fig 2.8 Petrol fuel system (carburettor type)

- Fuel tank (steel / plastic, baffles, vent, filler cap); fuel lines (steel / nylon); filters (sediment bowl, paper element 10-20 micron); pump; carburettor / injectors; air cleaner (paper, oil-bath).

Diesel fuel system (overview)

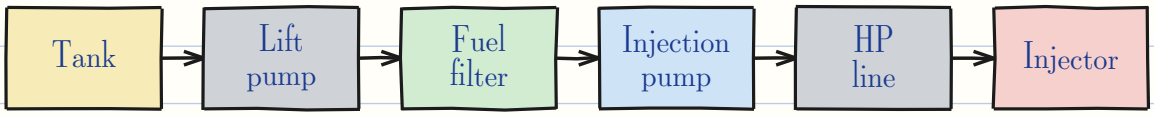


Fig 2.9 Diesel fuel system (inline pump type)

- Diesel = high pressure injection (150-2000 bar) of finely atomised fuel into the compressed air near TDC. Quantity governed by the governor / ECU.

Air-fuel ratio (A/F)

Condition	A/F (by mass)
Idling / cold start	10:1 - 12:1 (rich)
Cruising (economy)	15:1 - 17:1 (lean)
Full power	12:1 - 13:1
Stoichiometric (chemically correct)	14.7 : 1

9. Carburettor - Principle

- A carburettor atomises and vapourises petrol and mixes it with air in the proper ratio for different engine conditions.
- Principle: Bernoulli / venturi effect - air accelerates through a narrow throat (venturi) -> pressure drops (depression 0.05 - 0.07 bar) -> fuel is pushed from the float chamber through the jet into the air stream.

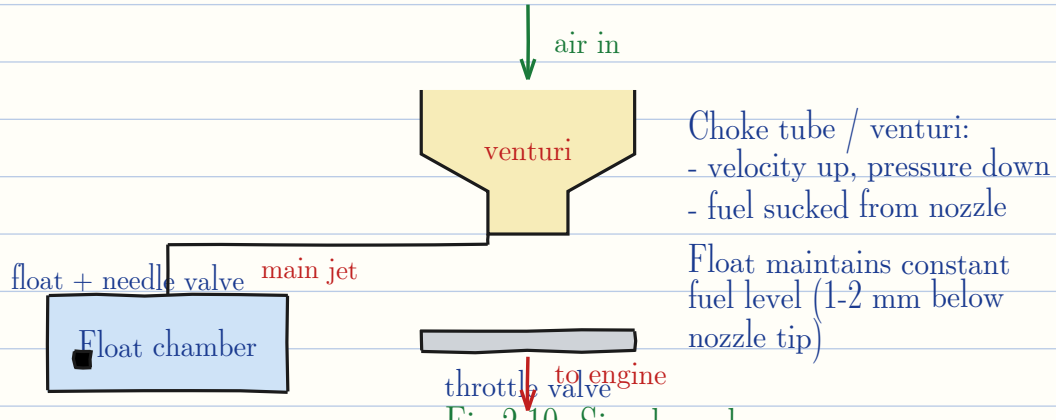


Fig 2.10 Simple carburettor

Parts of a simple carburettor

- Float chamber + float + needle valve (constant level); main jet / metering jet; discharge nozzle; venturi (choke tube); throttle valve (butterfly, controls quantity); choke valve (butterfly at air entrance).
- Defect of simple carburettor: gives increasingly rich mixture as throttle opens (flow rate grows) -> needs compensating devices (air bleed, compensating jet).

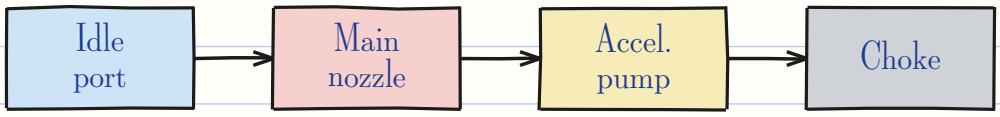
Requirements

Rich mixture for starting & idling, lean for cruising, slightly rich for full power; quick response in acceleration; good atomisation.

10. Carburettor Systems & Types

Practical carburettors (Solex, SU, Zenith, Carter) add circuits so that mixture suits every condition:

Circuit / device	Function
Choke system	Butterfly in air entry; closing it raises depression -> rich mixture for cold starting
Idling system	Idle jet + idle port near throttle edge; supplies rich mixture when throttle is almost closed
Main (cruising) system	Main jet + compensating / air-bleed jet keeps mixture lean (15:1) at medium throttle
Accelerator pump	Piston / diaphragm squirts extra petrol when throttle is opened suddenly (avoids flat spot)
Power enrichment (economiser)	Extra fuel via metering rod / vacuum valve at full load
Anti-percolation / vent	Prevents vapour lock and fuel overflow when hot



Each circuit takes fuel from the same float chamber

Fig 2.11 Circuits of a modern carburettor

Types of carburettors

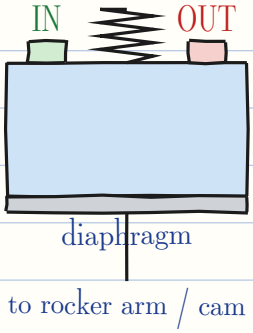
- By draught: updraught (air upward), downdraught (most common - gravity helps), horizontal / side-draught (SU, motorcycles).
- By venturi: fixed venturi (Solex, Zenith) vs constant-depression / variable-venturi (SU, CV - a piston varies the choke area; mixture almost constant).
- Drawbacks: poor atomisation / distribution to cylinders, icing, no accurate A/F control -> replaced by injection (BS-III / BS-IV emission norms).

11. Fuel Pumps

- Pump draws fuel from the tank and delivers it to the carburettor (0.2 - 0.3 bar) or injection rail (3 - 6 bar for MPFI, 100-200 bar for GDI).

1. Mechanical (diaphragm) pump

- Driven by an eccentric on camshaft -> rocker arm -> pulls diaphragm down (suction, inlet valve opens) and spring pushes it up (delivery, outlet valve opens).
- Delivery pressure is limited by the diaphragm spring - when carburettor needle valve closes, the diaphragm stays down (no flow) => self-regulating.



- Suction stroke: diaphragm pulled down, inlet valve opens, fuel enters chamber
- Delivery stroke: spring pushes up, outlet valve opens, fuel to carburettor

Fig 2.12 Mechanical diaphragm fuel pump

2. Electric fuel pump

- Plunger (SU) type - solenoid + contact breaker moves a diaphragm/bellows; mounted near tank. Rotary types: roller-cell, gear (gerotor), turbine (in-tank pump - quiet, cool, prevents vapour lock).
- Used in EFI / MPFI - provides constant high pressure; relay cut-off if engine stops (safety). Delivery 100-150 l/h at 3-5 bar.

3. High-pressure pumps

- GDI: engine-driven piston pump (100-200 bar). Diesel: inline / rotary / common-rail pump (1000-2000 bar). Lift (feed) pump supplies the HP pump.

12. Petrol Fuel Injection - Mono & Multi Point

- Fuel injection = fuel is sprayed under pressure into the intake air by injectors (solenoid valves) controlled by an ECU, instead of being sucked by a carburettor.

Advantages over carburettor

- Accurate A/F ratio for each cylinder; better power, mileage & emissions; quick throttle response; no icing; easy cold start; better cylinder-to-cylinder distribution.

Classification

- Mono-point / throttle-body injection (TBI, SPI): one or two injectors above the throttle valve - low cost, uses existing manifold; mixture still travels through the manifold (wall wetting).
- Multi-point / port injection (MPFI): one injector for each cylinder, near intake valve in the port -> precise & fast. Modes: simultaneous (all fire together), group (banks), sequential (each in firing order, best).
- Direct injection (GDI): injector inside combustion chamber.

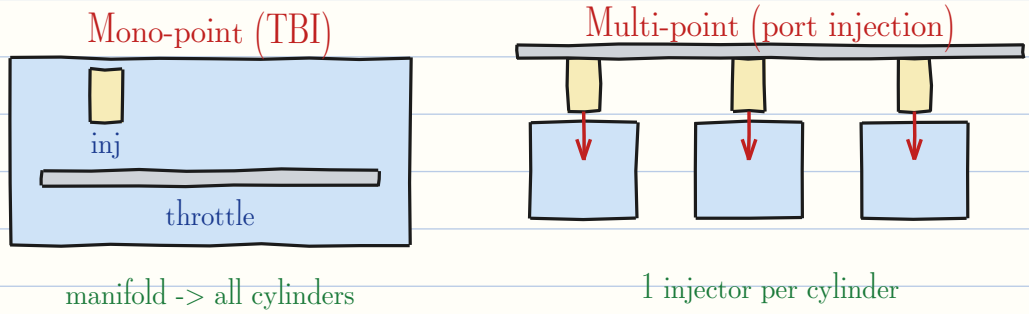


Fig 2.13 Mono-point vs multi-point injection

Point	Mono-point (TBI)	Multi-point (MPFI)
No. of injectors	1 (or 2)	One per cylinder
Mixture control	Fair	Excellent
Cost	Low	Higher
Fuel pressure	~1 bar	3 - 4 bar

13. Diesel Injection & Unit Injector

- Diesel engines need high-pressure, precisely timed and metered injection into the compressed air. Systems:

1. Inline (jerk) pump system - one plunger-barrel element per cylinder driven by a camshaft; quantity controlled by rotating the plunger (helical groove) through a rack; governor controls rack. Heavy trucks.
2. Distributor (rotary) pump - one pumping plunger + distributor rotor serves all cylinders (Bosch VE, CAV DPA). Compact; cars & LCVs.
3. Common rail direct injection (CRDI) - high-pressure pump (1600-2000 bar) fills a common rail; ECU opens solenoid / piezo injectors at any time -> pilot, main & post injection. Quiet, clean, efficient.
4. Unit injector (UI / EUI) and unit pump - pump + injector combined in one unit per cylinder, driven by camshaft rocker; no high-pressure lines (up to 2000-2200 bar).

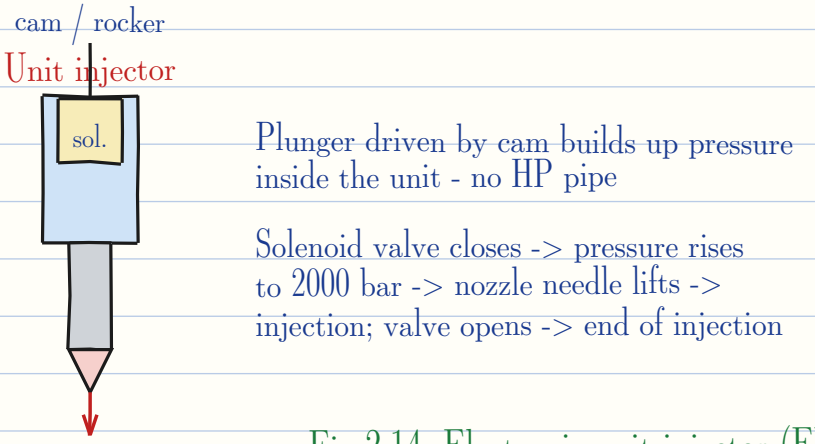


Fig 2.14 Electronic unit injector (EUI)

- Advantages of unit injector: very high pressure, fast response, precise timing by ECU, no pipe pressure waves. Used in Cummins, Caterpillar, Volvo, Detroit heavy-duty engines.
- Common-rail parts: HP pump, rail with pressure sensor & regulator, injectors, ECU, sensors.

14. Injector Nozzle Types

- The nozzle (injector) atomises fuel and distributes it in the combustion chamber. Needle valve is held on its seat by spring; opens when fuel pressure exceeds opening (popping) pressure: 120 - 250 bar (IDI: 100-150; DI: 200-300).
- Holder carries nozzle body, needle, spring, adjusting screw, leak-off (back-leak) line.

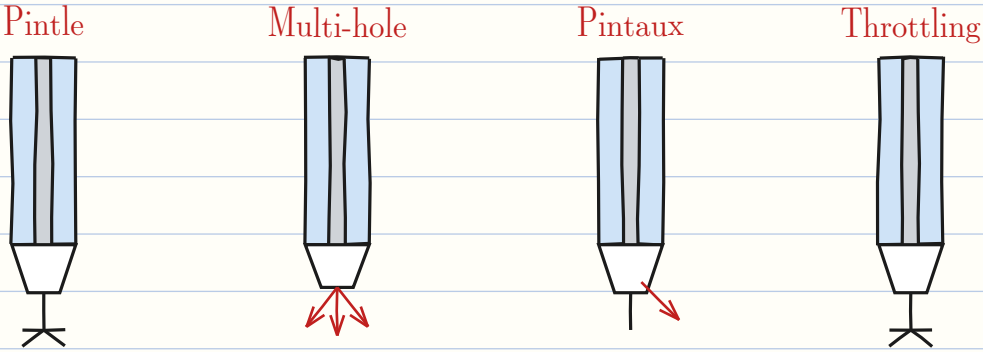


Fig 2.15 Types of diesel injector nozzles

Nozzle	Features / use
Single-hole	One central hole, 8-10 mm spray angle small; 175 bar; open chamber (DI) older engines
Multi-hole	4-8 holes (0.25-0.4 mm) at spray angle ~150 deg; direct injection engines; 200-300 bar
Pintle	Pin protrudes through orifice; spray cone 0-60 deg; self-cleaning; IDI / swirl chambers; 100-135 bar
Delay (throttling) pintle	Pin shaped to give small initial flow -> reduces diesel knock (IDI)
Pintaux	Pintle with auxiliary hole -> cold-start spray on to hot area; 3-hole variation

- Faults: nozzle choking (carbon), needle sticking, dripping, wrong opening pressure -> smoke, knock; tested on a nozzle tester.

15. Electronic Fuel Injection (EFI)

- EFI uses an ECU (electronic control unit) that reads sensors, calculates the required fuel quantity, and opens injectors for a calculated pulse width (1.5-10 ms) - closed-loop control with the lambda (O2) sensor.

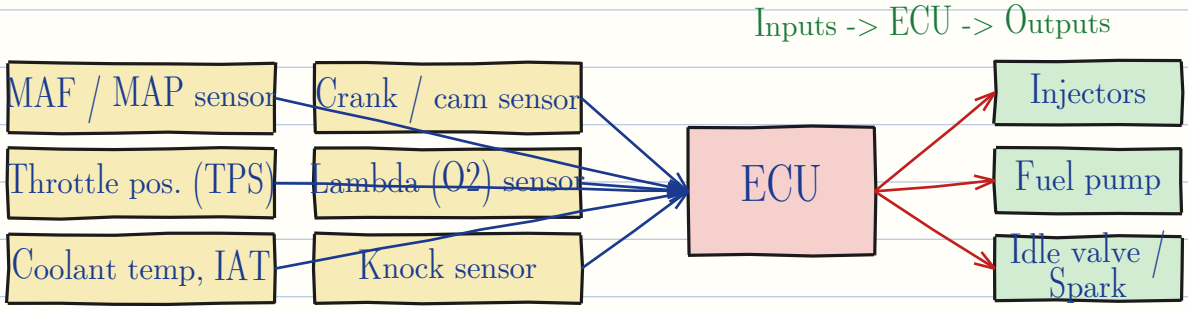


Fig 2.16 EFI system layout (block diagram)

Main sensors

- MAF (hot-wire air-mass), MAP (manifold pressure), TPS (throttle potentiometer), CTS / ECT (NTC thermistor), IAT, crank position (CKP) & cam (CMP), O2 (lambda), knock (piezo).

Actuators

- Solenoid injectors (12-16 ohm), fuel pump, idle air control / electronic throttle, ignition coils, EGR valve, canister purge valve.

Operating principle

- Basic pulse width = f(air mass, rpm) -> corrected for temperature, battery voltage, lambda feedback; modes: start, warm-up, idle, acceleration, full load enrichment, deceleration fuel cut-off.
- Closed loop keeps lambda = 1 so that the three-way catalytic converter works at best efficiency; OBD stores fault codes (DTC).

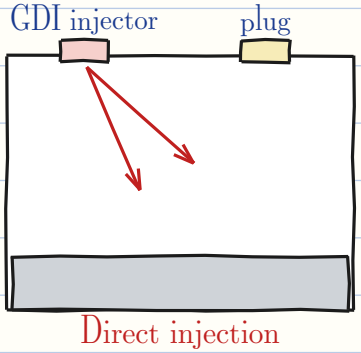
16. MPFI and GDI

MPFI (Multi-Point Fuel Injection)

- Each cylinder has its own injector in the intake port, spraying fuel toward the back of the intake valve; fuel rail at 2.5 - 4 bar.
- Sequential timing: injector fires once per cycle, during the intake stroke of that cylinder.
- Benefits: smooth idle, better mileage (~10 %), lower CO / HC, good cold start, fast response.

GDI (Gasoline Direct Injection)

- Petrol is injected directly into the combustion chamber at 50 - 250 bar by a high-pressure injector (like diesel); engine-driven HP pump + rail + ECU.



Modes:

1. Stratified charge (late injection, compression stroke) - ultra-lean
2. Homogeneous (early injection, intake stroke) - at full load
3. Homogeneous lean / stoich + EGR

Fig 2.17 Gasoline direct injection

- Advantages: 15-20 % better fuel economy, higher compression ratio (10-12), cooling effect of fuel evaporation (less knock), more power & torque, supports turbo-downsizing (TSI, EcoBoost, FSI, Skyactiv-G).
- Disadvantages: costly HP pump & injectors, carbon deposits on intake valves (no fuel wash), more particulate (PM) / NOx -> gasoline particulate filter (GPF), noise.

MPFI vs GDI

MPFI: fuel injected in port (3-4 bar), always homogeneous. GDI: fuel injected in cylinder (50-250 bar), stratified or homogeneous; more efficient but costlier.

17. DTS-i (Digital Twin Spark) & Revision

- DTS-i (Digital Twin Spark ignition) is a Bajaj technology (Pulsar) using two spark plugs per cylinder controlled digitally (CDI).
- Why two plugs? In a wide-bore, flat-piston, short-stroke engine one central plug cannot burn the mixture quickly: the flame has to travel far. Two plugs at opposite ends start two flame fronts that meet in the middle.

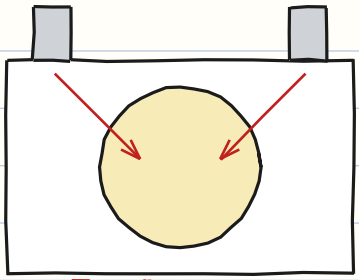


Fig 2.18 Twin spark plugs in combustion chamber

DTS-i family:
- DTS-i : 2 plugs + digital CDI (fast burn)
- DTS-Si : 2 plugs + swirl induction for better air-fuel mixing, higher mileage
- DTS-FI : DTS + electronic fuel injection

Advantages of DTS

- Faster, complete combustion -> more power, better mileage, lower emission; the ECU/CDI can retard / advance each plug independently; smooth running at low speed.
- Plugs may fire together or with a slight lag depending on rpm (map-controlled).

Quick revision - ignition & fuel

System	Key point
Coil ignition	Primary 12 V -> secondary 20-30 kV; CB + condenser
Magneto	Self-generating; two-wheelers
Spark plug	Gap 0.6-0.9 mm; hot / cold by heat range
Carburettor	Venturi principle; A/F 15:1 cruise
MPFI / EFI	ECU + sensors + injector pulse width
GDI	In-cylinder injection 50-250 bar